

(TR-104) PYTHON WITH AI/ML

Training Day 68 Report

Date: 15 April 2026

Topic Covered: AI Instagram Reel Generation, Text-to-Video Models & Stable Diffusion Video Pipelines

The sixty-eighth day of training focused on AI-powered Instagram Reel generation workflows using Tavily Search, Ollama, Hugging Face Diffusers, Stable Diffusion, and OpenCV. The training also included implementation of text-to-video generation pipelines, frame-based video generation techniques, and optimization strategies for CPU-based AI video workflows.

- Integrated Tavily Search API to fetch real-time trending Instagram reel content based on user-defined topics.
- Built a dynamic query system for retrieving trend-related data for Indian corporate Instagram reels.
- Processed and cleaned fetched search results to generate structured inputs for LLM workflows.
- Implemented Ollama (qwen2.5:3b) for generating professional 30-second Instagram reel scripts.
- Designed prompt engineering strategies for structured reel script generation with integrated “VIDEO PROMPT” extraction.
- Developed automated logic for extracting video prompts from generated LLM outputs.
- Generated professional Instagram captions using constrained prompt generation techniques.
- Implemented fallback mechanisms for handling incomplete or missing video prompts.
- Built a complete end-to-end automated workflow integrating trend fetching, script generation, caption generation, and video creation.
- Implemented text-to-video generation using damo-vilab/text-to-video-ms-1.7b model from Hugging Face Diffusers.
- Configured the video generation pipeline to run entirely on CPU-based systems.

- Applied memory optimization techniques such as attention slicing for efficient CPU execution.
- Generated AI-powered videos from text prompts with configurable inference settings.
- Exported generated frame sequences into .mp4 video format using export_to_video utility.
- Handled performance limitations and long inference times in CPU-based text-to-video workflows.
- Implemented Stable Diffusion pipeline using runwayml/stable-diffusion-v1-5 for high-quality frame generation.
- Generated multiple consistent video frames using fixed random seed for reproducibility.
- Converted generated images into NumPy arrays for OpenCV-based processing.
- Applied frame blending techniques for smoother transitions between generated frames.
- Designed a custom intermediate frame generation pipeline for improving video smoothness and visual quality.
- Used OpenCV VideoWriter to combine generated frames into structured video output.
- Configured FPS and video resolution settings for smoother and visually consistent results.
- Successfully improved generated video quality compared to basic frame stitching methods.
- Reviewed poster content for spelling, grammar, and presentation quality improvements.
- **GitHub Link:**
https://github.com/JaspinderKaurWalia26/video_generation
- **Blog Link:**
<https://medium.com/@jaspinderkaurwalia855/turning-words-into-motion-ai-video-generation-on-a-simple-cpu-2e9b555bfb78>

Overall, the training strengthened understanding of AI-powered video generation, Hugging Face Diffusers, Stable Diffusion workflows, OpenCV-based frame processing, prompt engineering, and automated Instagram Reel generation systems.